SECTION 4
PERFORMANCE DATA

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SECTION 4
PERFORMANCE DATA

INTRODUCTION

The performance data presented herein is derived from the engine manufacturer's specification power for the engine less installation losses. This data is applicable to the basic helicopter without any optional equipment which would appreciably affect lift, drag, or power available.

NOTE

Data shown at -30°C are provided to allow interpolation at -25°C.

POWER ASSURANCE CHECKS

The Power Assurance Check charts are provided in order to let the pilot determine if the engines can produce the installed specification power.
A power assurance check should be performed daily. Additional checks should be made if unusual operating conditions or indications arise.
The hover check is performed prior to takeoff and the in-flight check is provided for periodic in-flight monitoring of engine performance.
Either power assurance check method may be selected at the discretion of the pilot. It is pilot's responsibility to accomplish the procedure safely, considering passenger load, terrain being overflown and the qualification of persons on board to assist in watching for other air traffic and to record power check data.
If either one of the two engines does not meet the requirements of the hover or the in-flight check, the minimum performance requirements as per published data may not be obtained.
The cause of engine power loss, or excessive TOT or GAS PRODUCER RPM (N1) should be determined as soon as practical.
Refer to Engine Maintenance Manual.

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POWER ASSURANCE CHECK
HOVER

* HEATER / E.C.S. OFF.
* GENERATOR LOAD TO MINIMUM (< 28 A).
* SET NR TO 102%.
* TEST ENGINE MODE SWITCH - FLIGHT.
* OTHER ENGINE MODE SWITCH - IDLE.

* INCREASE COLLECTIVE UNTIL LIGHT ON WHEELS OR HOVERING AT 3 FEET, DO NOT EXCEED 820°C TOT OR 97.4% N1 OR 124% TORQUE.
* STABILIZE POWER ONE MINUTE, THEN RECORD OAT, PRESSURE ALTITUDE, ENGINE TORQUE, TOT AND N1.
* ENTER CHART AT INDICATED ENGINE TORQUE, MOVE DOWN TO INTERSECT PRESSURE ALTITUDE, PROCEED TO THE RIGHT TO INTERSECT OUTSIDE AIR TEMPERATURE, THEN MOVE UP TO READ VALUES FOR MAXIMUM ALLOWABLE TOT AND GAS PRODUCER RPM (N1).
* IF INDICATED TOT OR GAS PRODUCER RPM (N1) EXCEEDS MAXIMUM ALLOWABLE, REPEAT CHECK, STABILIZING POWER FOUR MINUTES.
* REPEAT CHECK USING OTHER ENGINE.
* IF EITHER ENGINE EXCEEDS ALLOWABLE TOT OR GAS PRODUCER RPM (N1) AFTER STABILIZING FOUR MINUTES, PUBLISHED PERFORMANCE MAY NOT BE ACHIEVABLE. CAUSE SHOULD BE DETERMINATED AS SOON AS PRACTICAL.

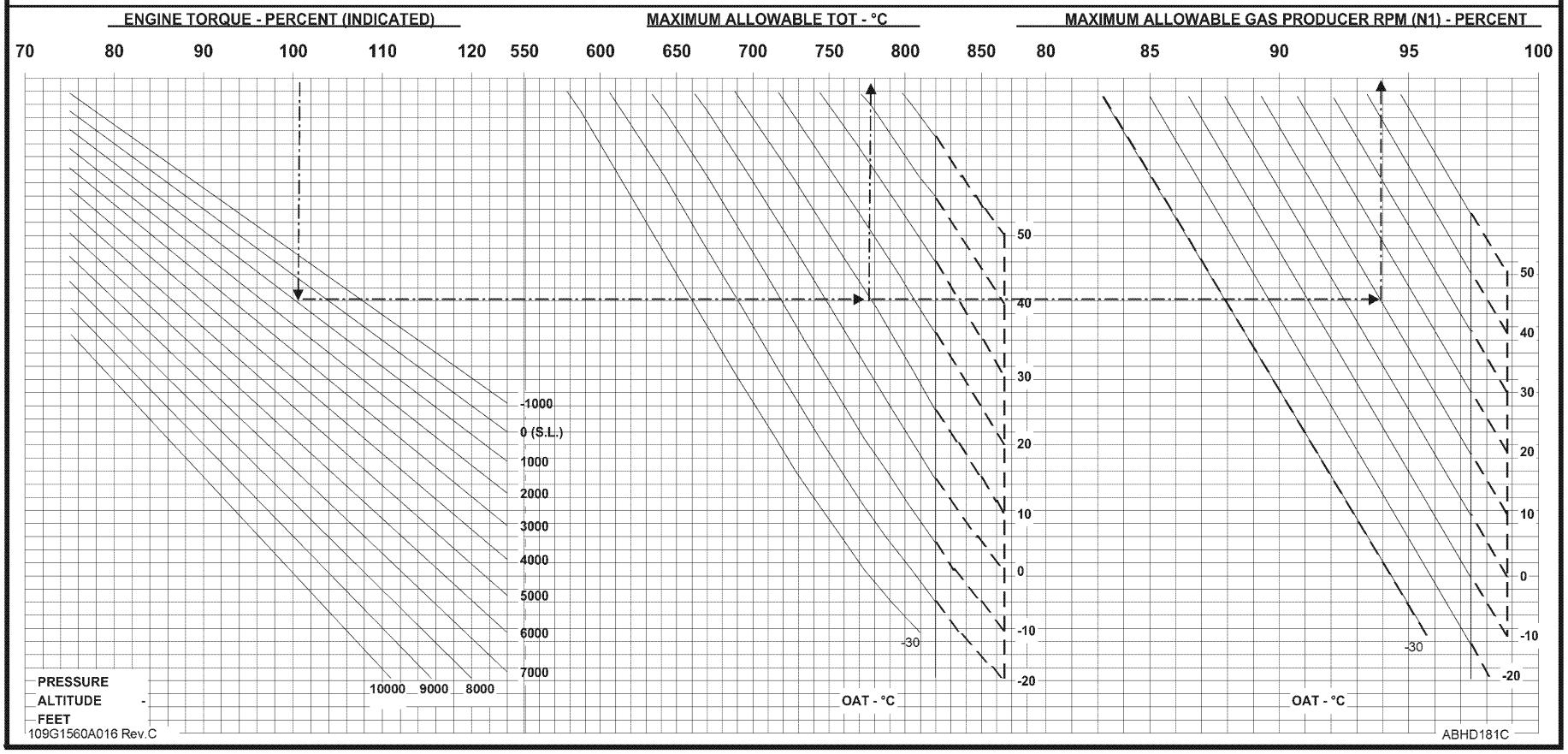

Figure 4-1. Power assurance check - hover.

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POWER ASSURANCE CHECK
IN FLIGHT

* HEATER / E.C.S. OFF.
* GENERATOR LOAD TO MINIMUM (< 28 A).
* SET NR TO 100%.
* ESTABILISH LEVEL FLIGHT ABOVE 1000 ft. AGL.
* AIRSPEED - 100 KIAS.
* TEST ENGINE - SET ENG. GOV. SWITCH TO MANUAL
* OTHER ENGINE - LEAVE ENG. GOV. SWITCH TO AUTO

* OPERATE TEST ENG TRIM TO INCREASE POWER UNTIL ENGINE TORQUE IS WITHIN TEST RANGE. DO NOT EXCEED 820°C TOT OR 97.4% N1 OR 100% TORQUE.
* STABILIZE POWER ONE MINUTE IN LEVEL FLIGHT THEN RECORD OAT, PRESSURE ALTITUDE, ENGINE TORQUE, TOT AND N1.
* ENTER CHART AT INDICATED ENGINE TORQUE, MOVE DOWN TO INTERSECT PRESSURE ALTITUDE, PROCEED TO THE RIGHT TO INTERSECT OUTSIDE AIR TEMPERATURE, THEN MOVE UP TO READ VALUES FOR MAXIMUM ALLOWABLE TOT AND GAS PRODUCER RPM (N1).
* IF INDICATED TOT OR GAS PRODUCER RPM (N1) EXCEEDS MAX ALLOWABLE, REPEAT CHECK, STABILIZING POWER FOUR MINUTES.
* REPEAT CHECK USING OTHER ENGINE.
* IF EITHER ENGINE EXCEEDS ALLOWABLE TOT OR GAS PRODUCER RPM (N1) AFTER STABILIZING FOUR MINUTES, CARRY OUT A POWER ASSURANCE CHECK IN HOVER.

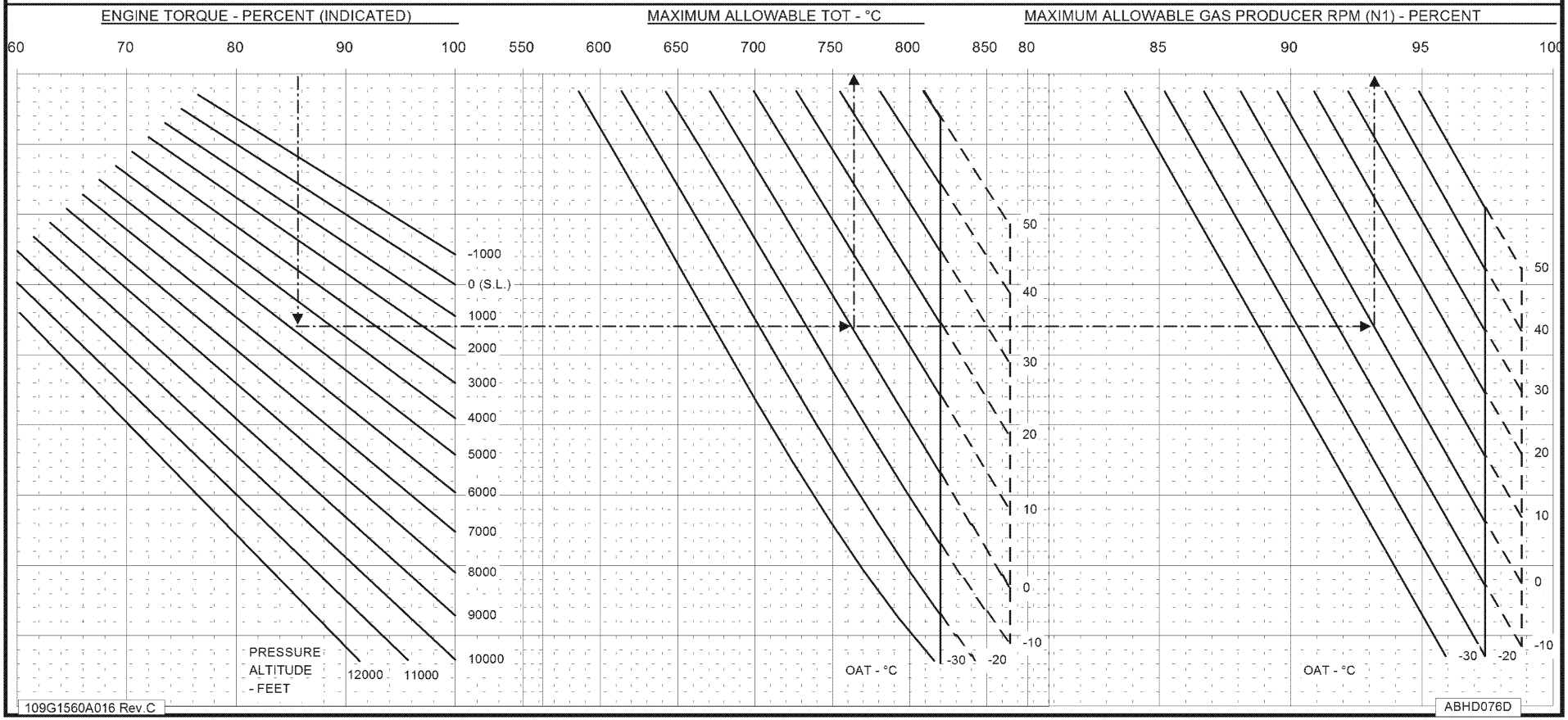

Figure 4-2. Power assurance check - in flight.

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DENSITY ALTITUDE

A Density Altitude Chart is provided to aid in calculation of performance and limitations.
Density altitude (Hd) is an expression of the density of the air on terms of height above sea level; therefore, the less dense the air, the higher the density altitude.
For standard conditions of temperature and pressure, density altitude is the same as pressure altitude (Hp).
As temperature increases above standard for any altitude, the density altitude will also increase to values higher then pressure altitude.
The chart shows density altitude as a function of pressure altitude and temperature.
The chart also enable to obtain the inverse of the square root of the density ratio ($^{1}/_{\sqrt{\sigma}}$), which is used to calculate the True Airspeed (TAS) by the relation:

$$TAS = CAS \times {}^{1}/_{\sqrt{\sigma}}$$

where CAS is the Calibrated Airspeed.

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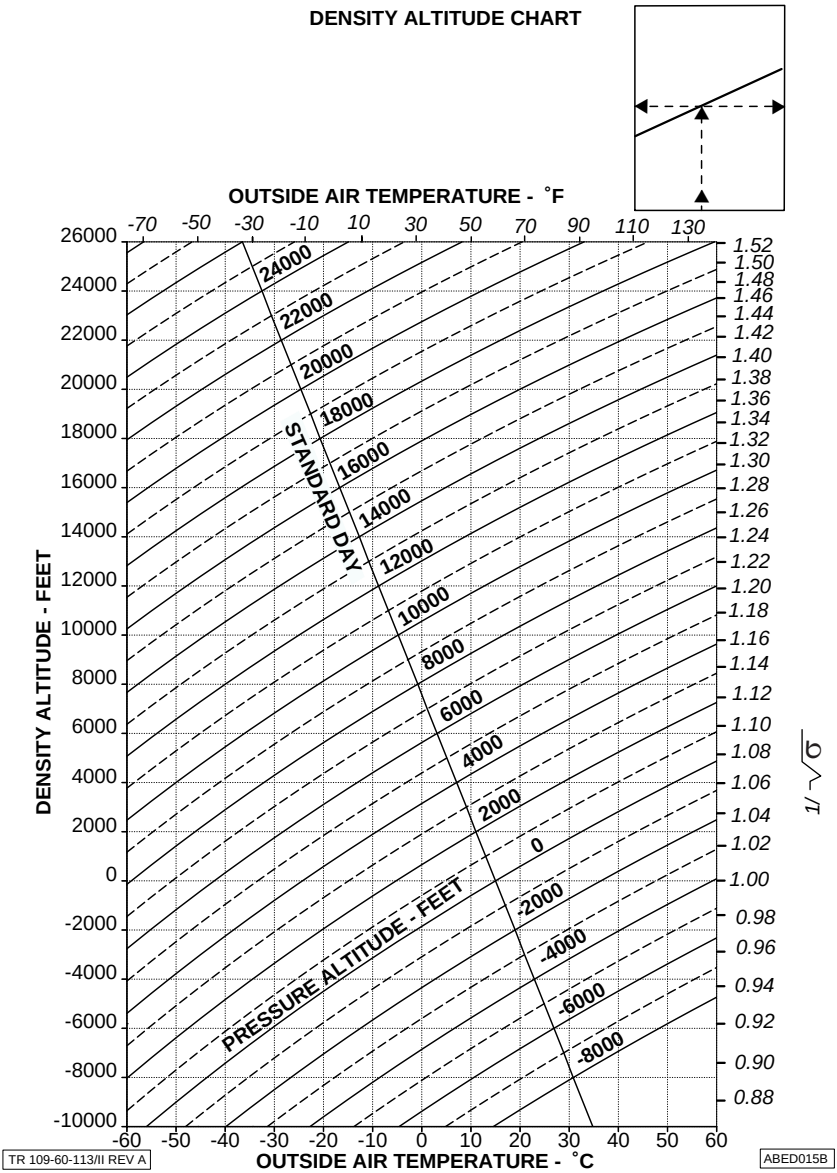

Figure 4-3. Density - altitude chart.

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CONVERSION CHART

A Conversion Chart is also provided for altitude, weight, temperature, airspeed, rate of climb and capacity.

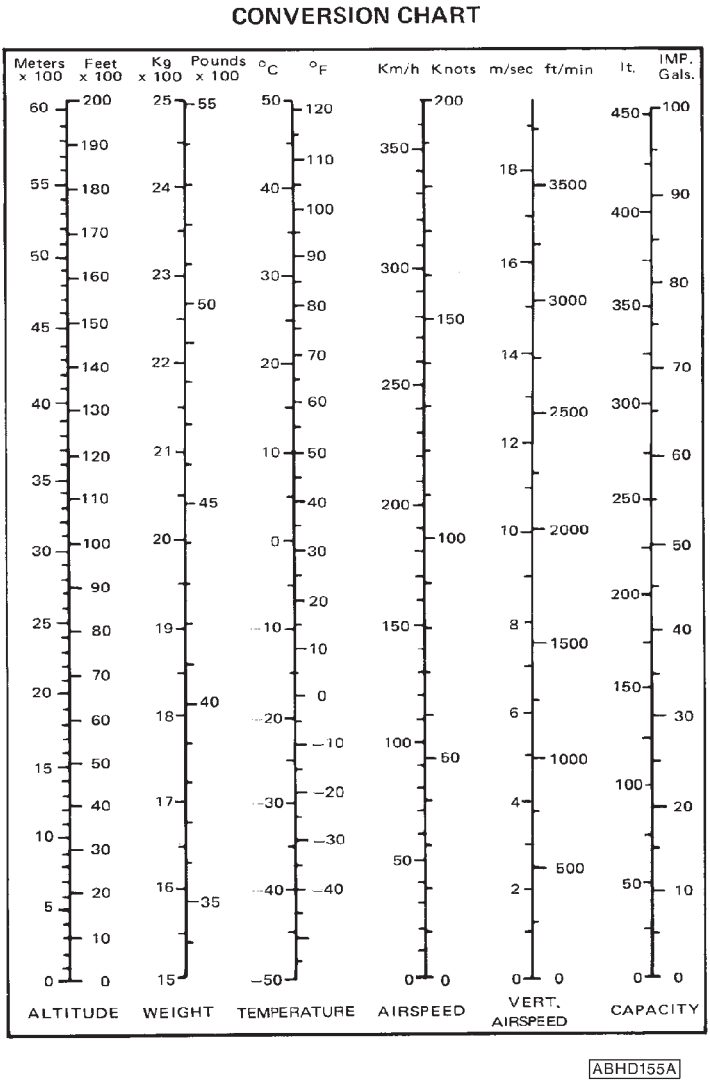

Figure 4-4. Conversion chart.

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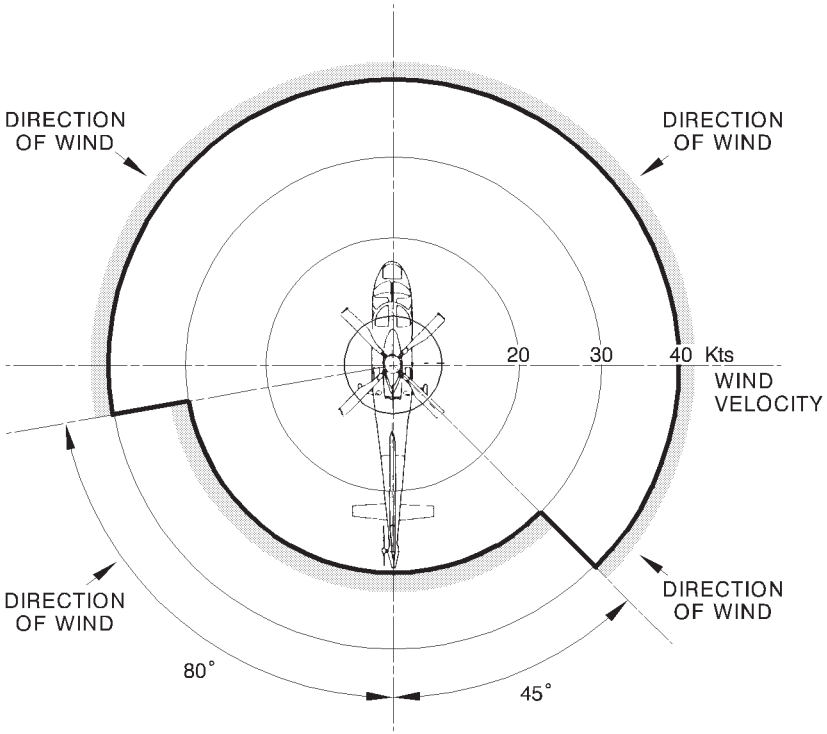
OPERATION VS ALLOWABLE WIND

Satisfactory stability and control in rearward and sideward flight has been demonstrated, at all loading conditions for hover in ground effect up to takeoff power, from -1000 ft to 8000 ft Hd, in the following wind/ground speed azimuth envelope:



APPLICABILITY: UP TO 4000 ft Hd

ABHD156A

Figure 4-5 (Sheet 1 of 2). Wind/ground speed azimuth envelope.

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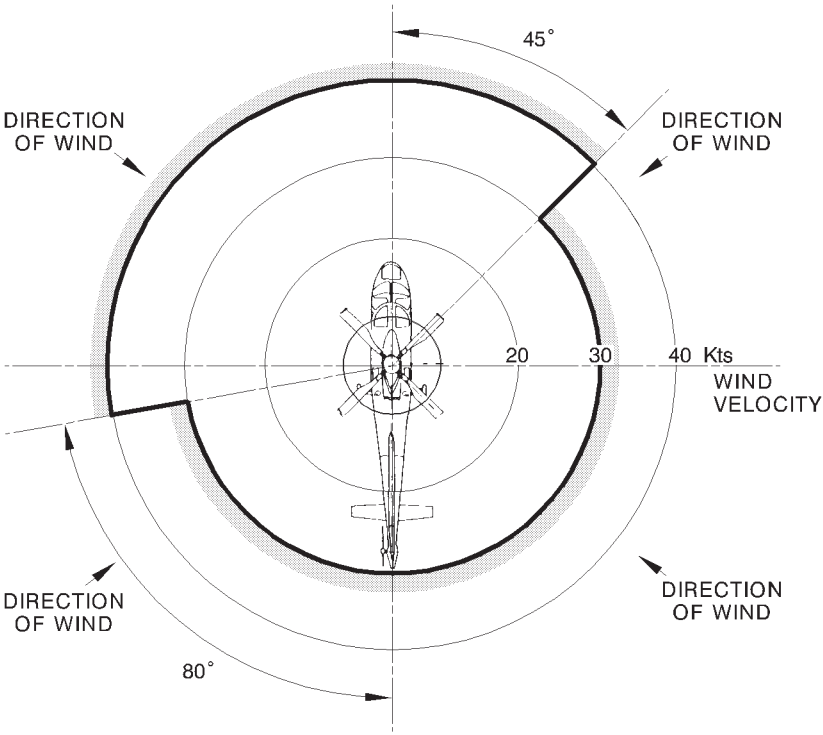

APPLICABILITY: FROM 4000 TO 8000 ft Hd

ABHD090C

Figure 4-5 (Sheet 2 of 2). Wind/ground speed azimuth envelope.

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HOVERING CEILING IGE

The Hovering Ceiling In Ground Effect charts provide the maximum gross weight for hovering IGE (3 ft above ground level) at all pressure altitudes and outside air temperatures, with main rotor speed at 102% and zero wind condition.
The charts are presented for takeoff power rating and for maximum continuous power rating.

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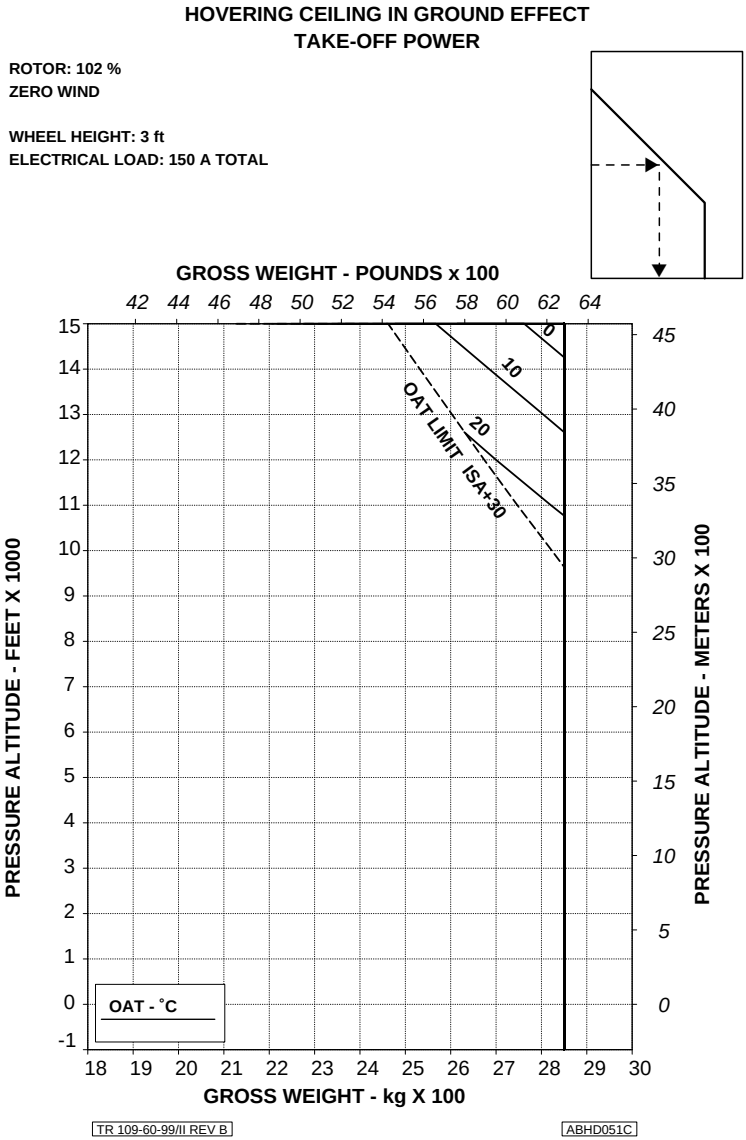

Figure 4-6. Hovering ceiling - in ground effect - take-off power.

FOR TRAINING ONLY

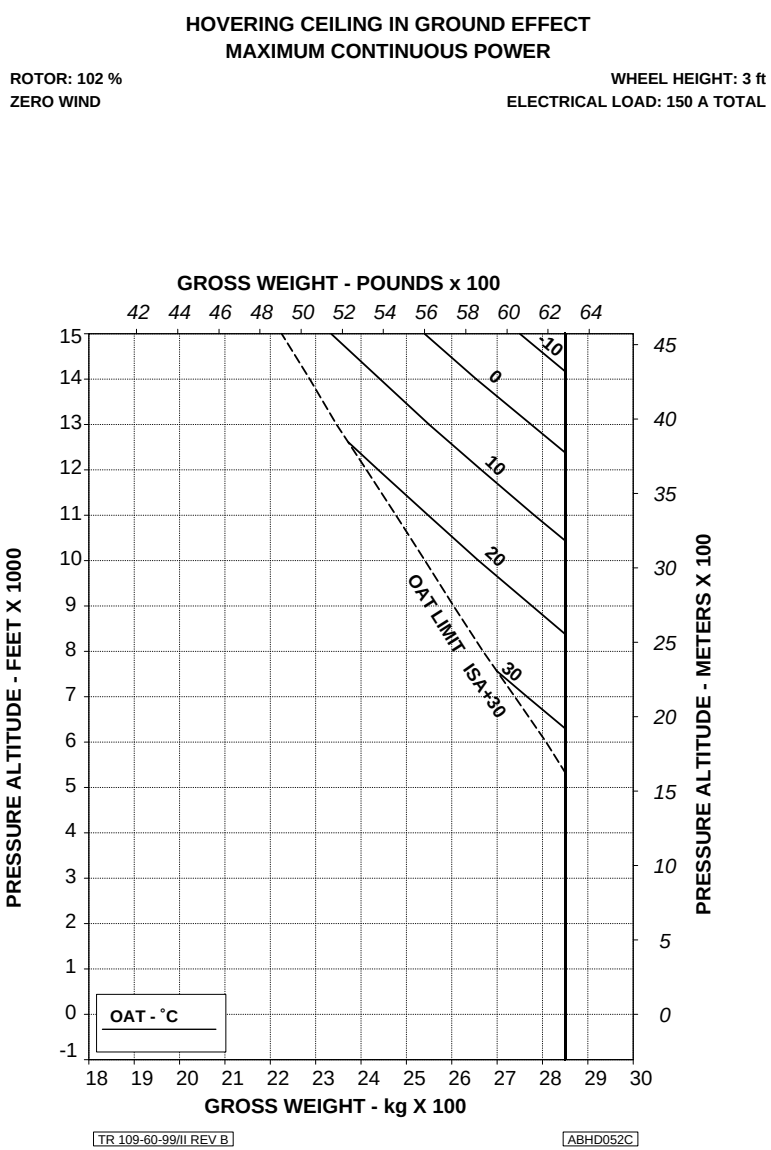

Figure 4-7. Hovering ceiling - in ground effect - maximum continuous power.

FOR TRAINING ONLY

HOVERING CEILING OGE

The Hovering Ceiling Out of Ground Effect charts provide the maximum gross weight for hovering OGE (60 ft above ground level) at all pressure altitudes and outside air temperatures, with main rotor speed at 102% and zero wind condition.
The charts are presented for takeoff power rating and for maximum continuous power rating.

FOR TRAINING ONLY

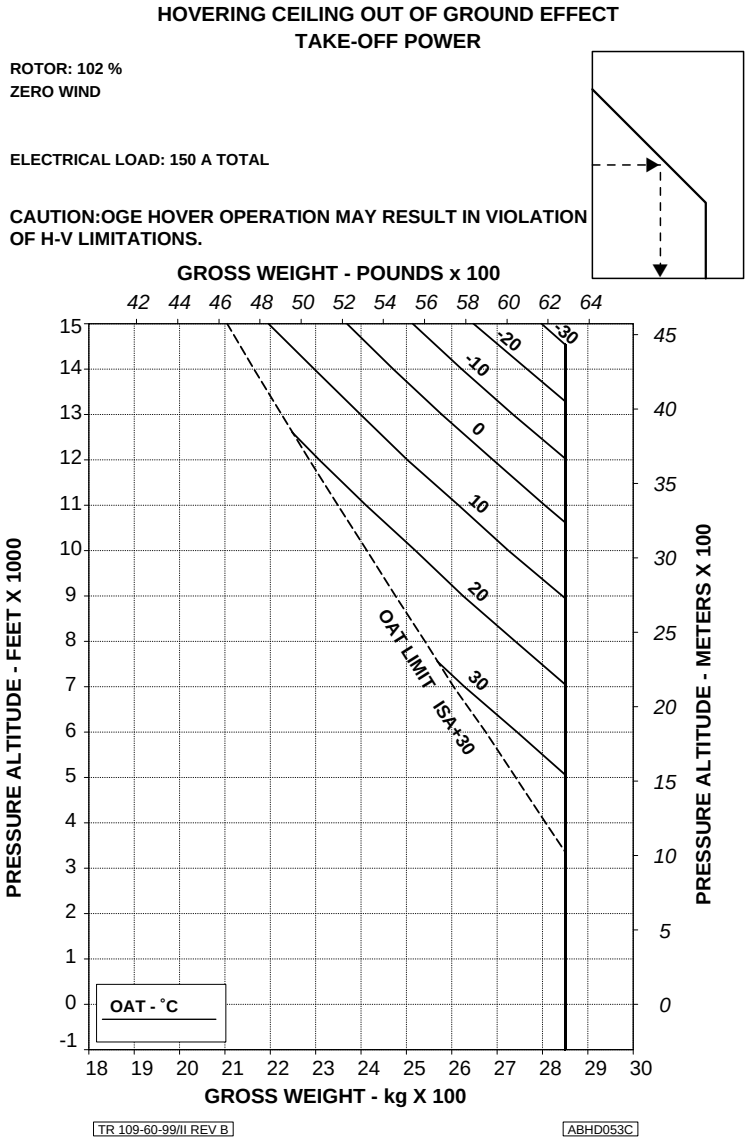

Figure 4-8. Hovering ceiling - out of ground effect - take-off power.

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HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
ZERO WIND **ELECTRICAL LOAD: 150 A TOTAL**

CAUTION:OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

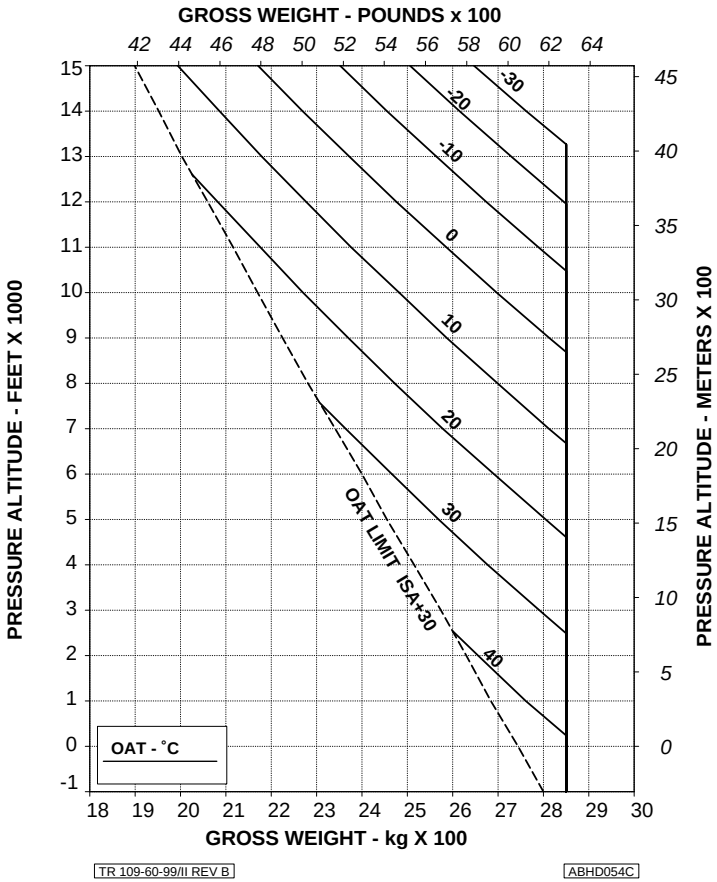

Figure 4-9. Hovering ceiling - out of ground effect - maximum continuous power.

FOR TRAINING ONLY

HEIGHT - VELOCITY DIAGRAM

The Height - Velocity diagram enable to establish if, in the event of a single engine failure during takeoff, landing or other operation near the surface, a combination of airspeed and height above ground exists from which a safe single engine landing on a smooth, level and hard surface cannot be assured (dangerous zone).
The height - Velocity diagram is split in two charts.
Chart A shows the weight values, together with outside air temperature and altitude, at/below which the dangerous zone does not exist. For heavier weights refer to Chart B.
Chart B defines the combinations of height and airspeed to avoid for safe operations.

NOTE

The height - velocity diagram does not define the conditions which assure continued flight following an engine failure nor the conditions from which a safe power off landing can be made.

FOR TRAINING ONLY

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

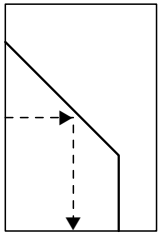

CHART "A" SHOWS THE WEIGHT VALUES AT/BELOW WHICH THERE IS NO H-V LIMITATION FOR HEAVIER WEIGHTS, REFER TO CHART "B".

CHART A

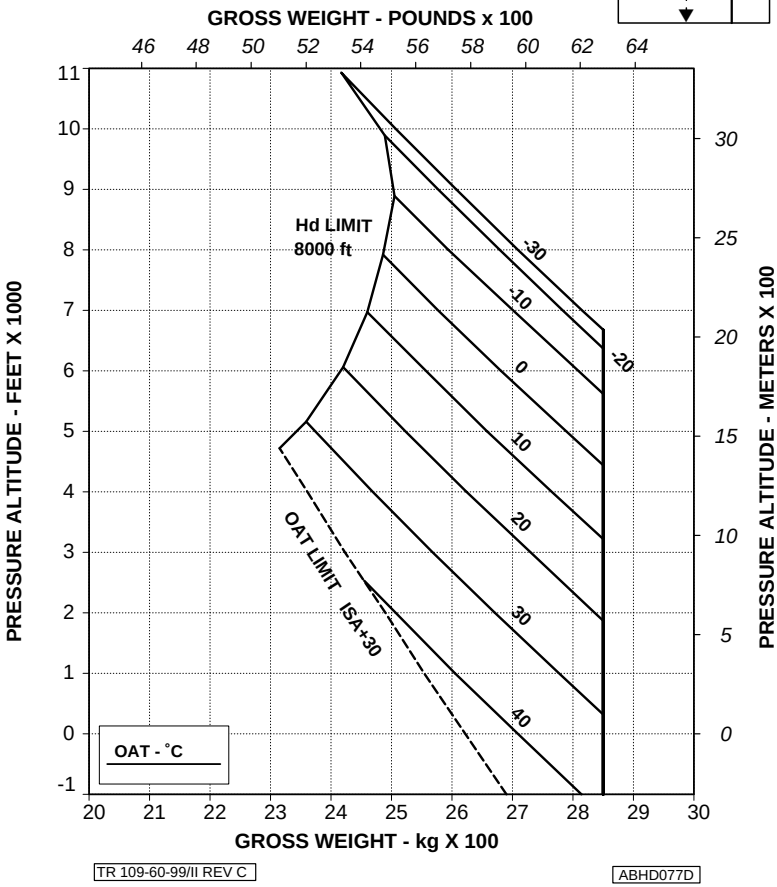

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Figure 4-10 (sheet 1 of 2). Height - velocity diagram.

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HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

FOR SMOOTH, LEVEL, HARD SURFACES

CHART B

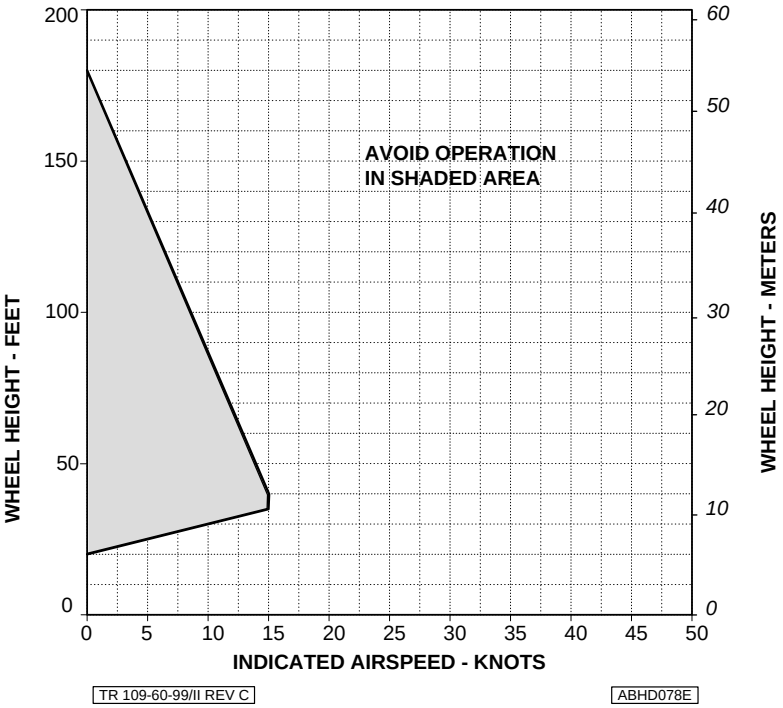

Figure 4-10 (sheet 2 of 2). Height - velocity diagram.

FOR TRAINING ONLY

RATE OF CLIMB

The Rate of Climb charts are provided with all engines operative and with one engine inoperative. The rate of climb AEO charts are presented for takeoff power rating and for maximum continuous power rating, both with main rotor speed at 100%, and they refer to the best rate of climb speed of 60 Kts IAS.The rate of climb OEI charts are presented for 2.5 minutes power rating and for maximum continuous power rating, both with main rotor speed at 100%, and they refer to the best rate of climb speed of 60 Kts IAS.

NOTE

Single engine performance is intended for emergency use when one engine becomes inoperative due to an actual malfunction and for maintenance or training purpose.
For maintenance and training OEI operation shall be limited to the maximum continuos OEI power rating.

FOR TRAINING ONLY

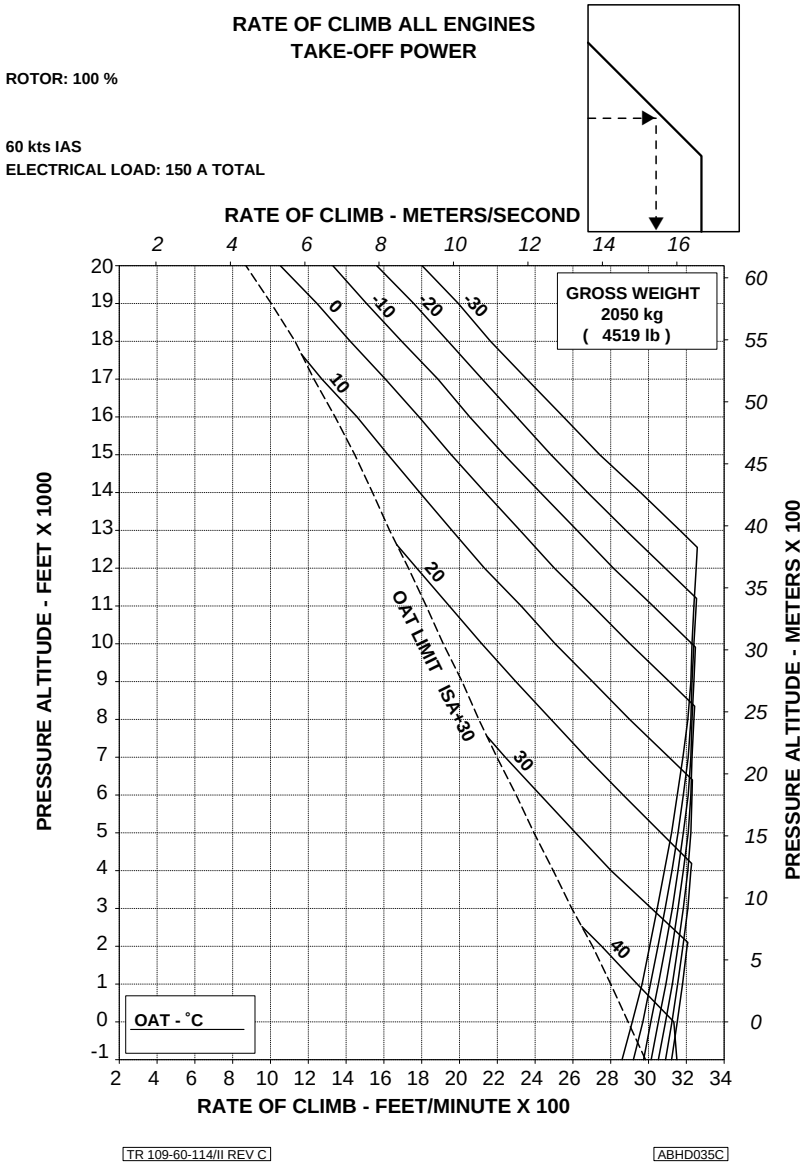

Figure 4-11. Rate of climb - all engines - take-off power - 2050 kg.

FOR TRAINING ONLY

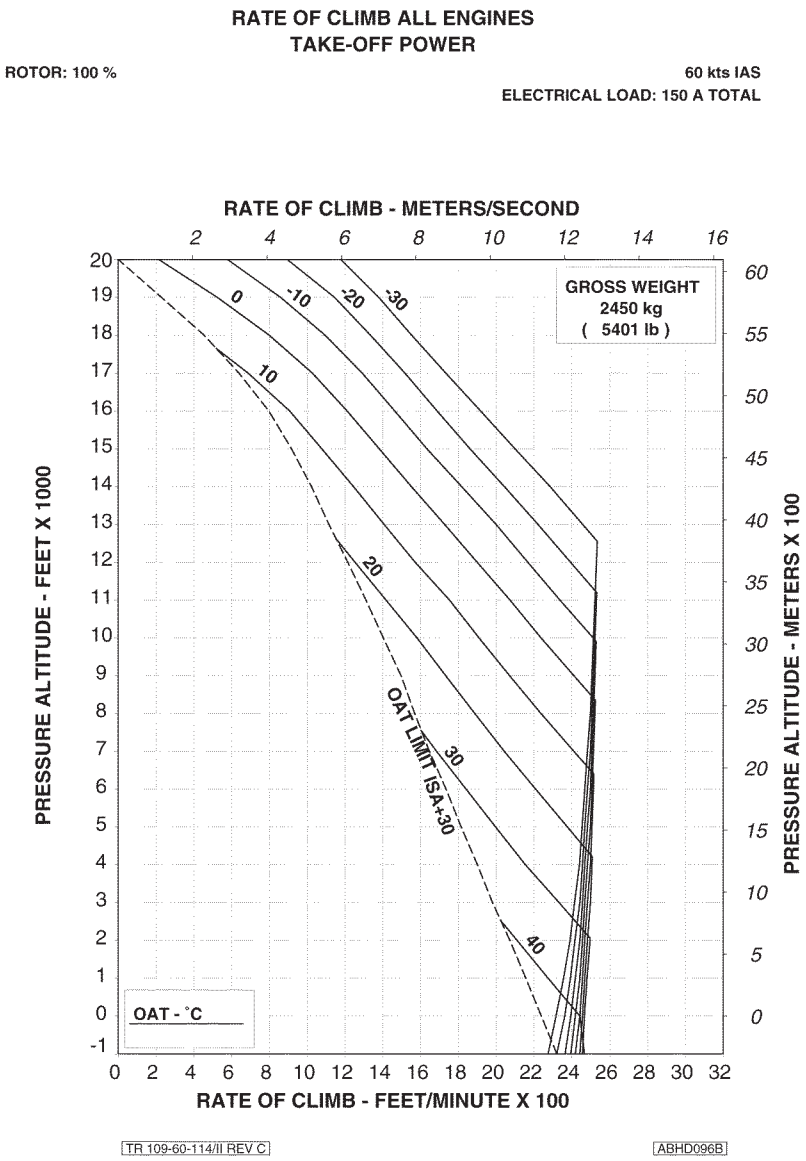

Figure 4-12. Rate of climb - all engines - take-off power - 2450 kg.

FOR TRAINING ONLY

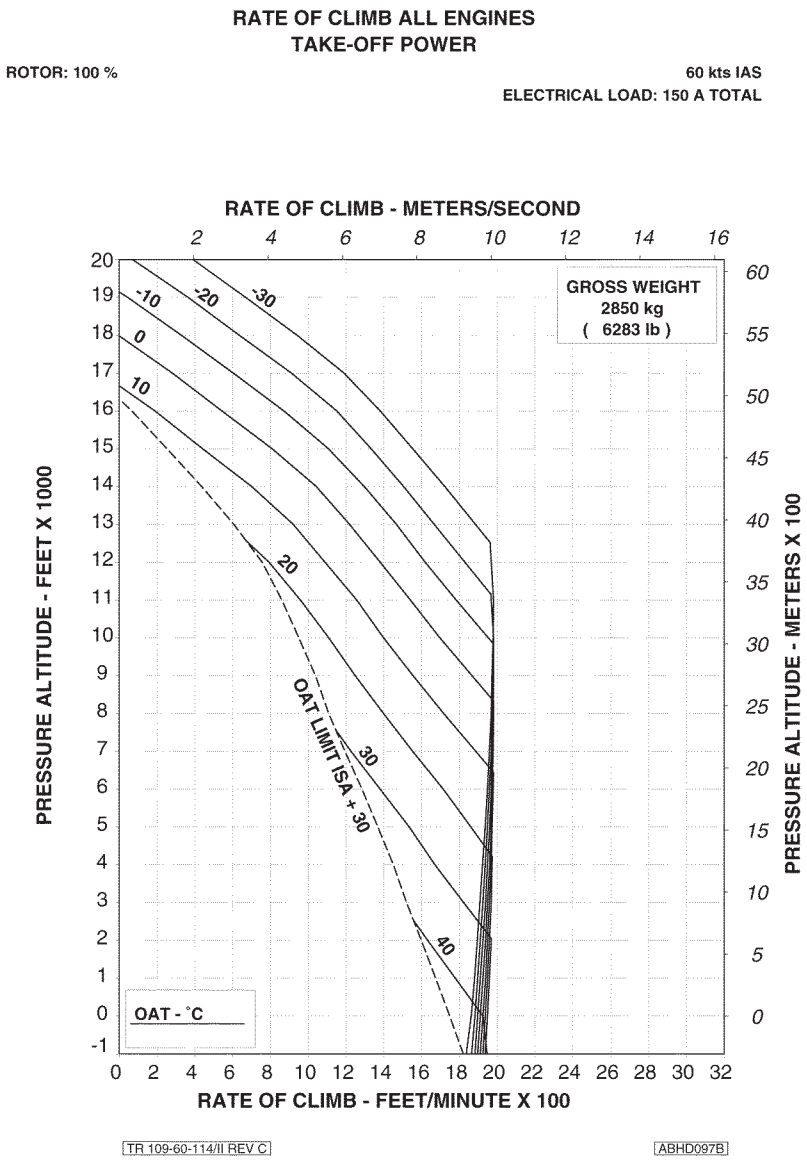

Figure 4-13. Rate of climb - all engines - take-off power - 2850 kg.

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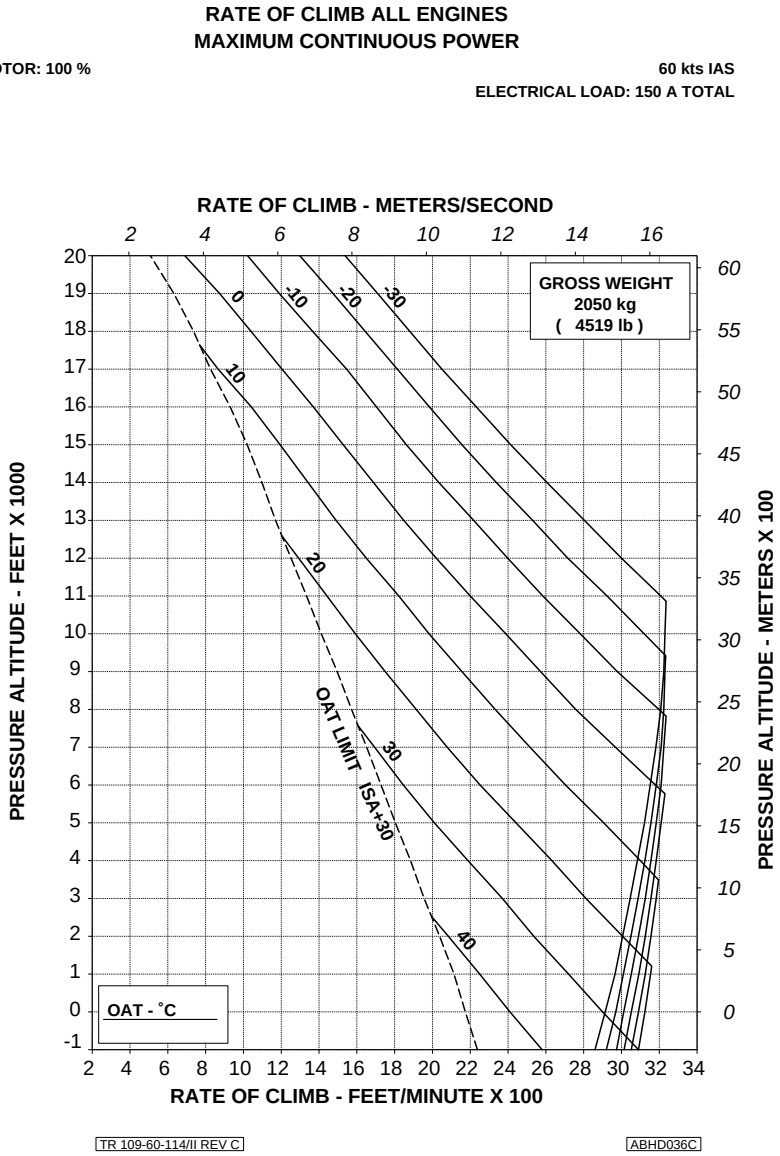

Figure 4-14. Rate of climb - all engines - maximum continuous power - 2050 kg.

FOR TRAINING ONLY

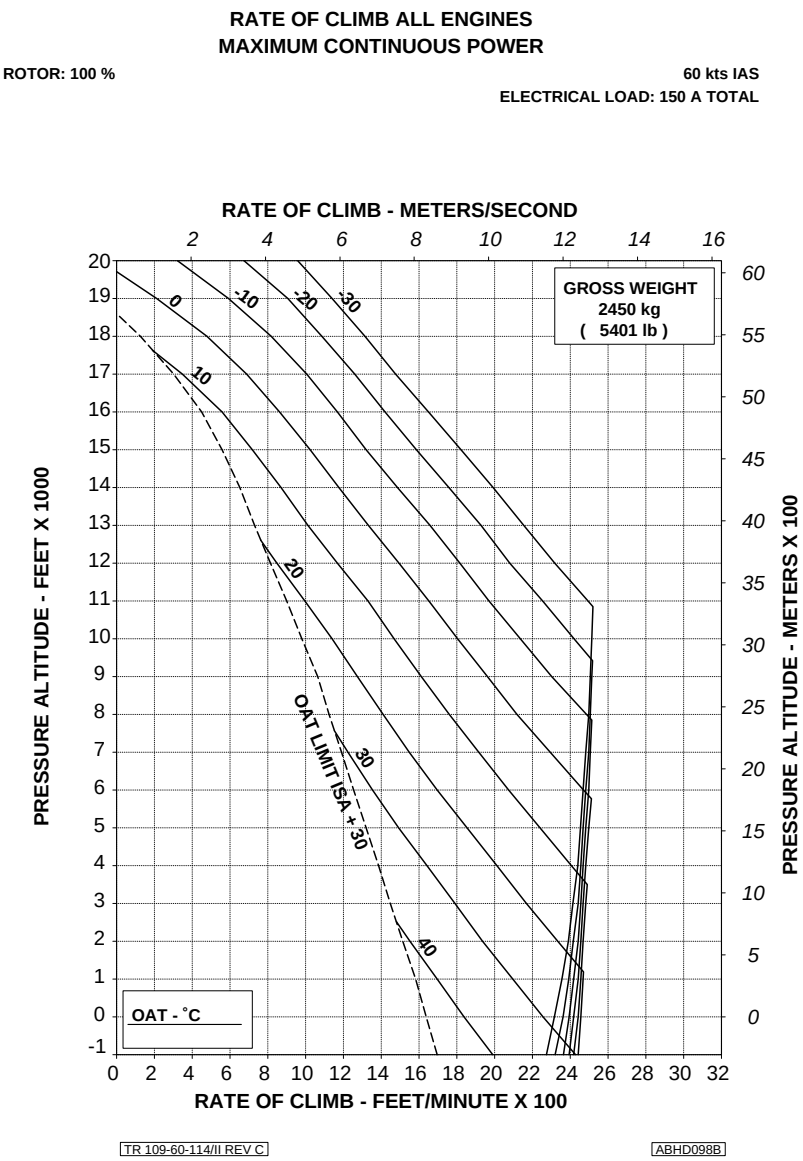

Figure 4-15. Rate of climb - all engines - maximum continuous power - 2450 kg.

FOR TRAINING ONLY

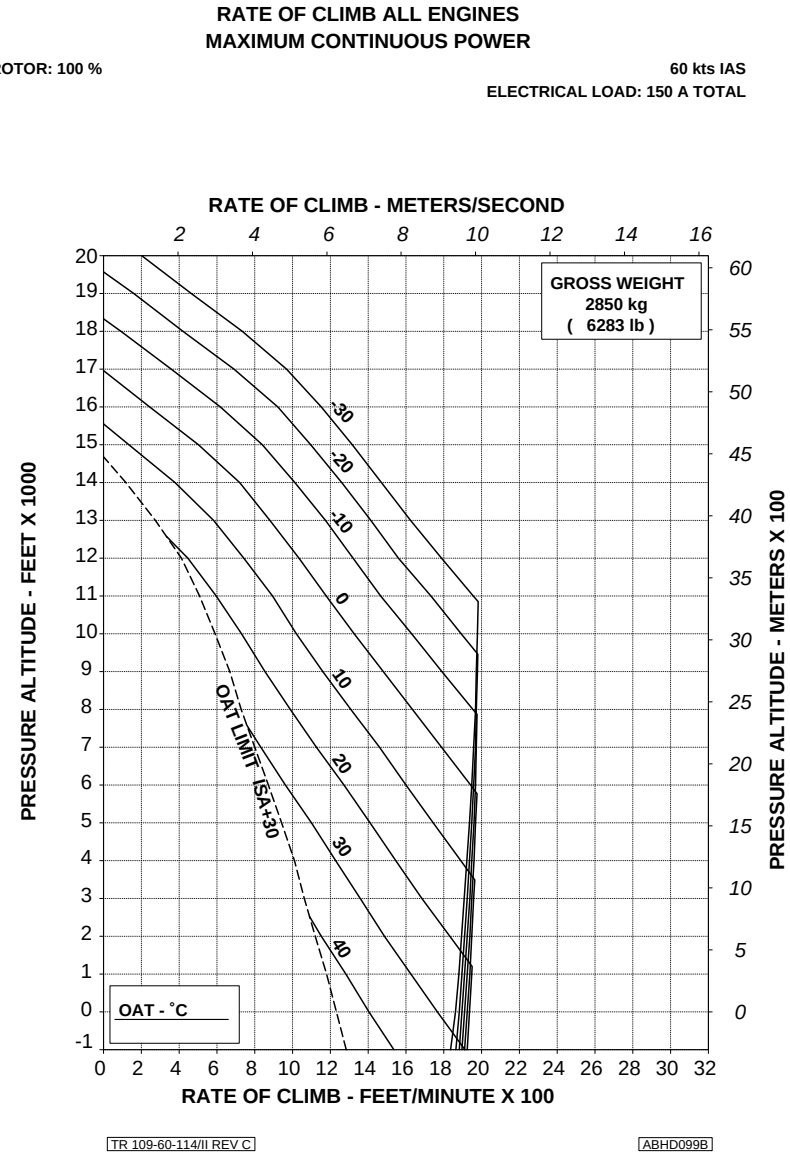

Figure 4-16. Rate of climb - all engines - maximum continuous power - 2850 kg.

FOR TRAINING ONLY

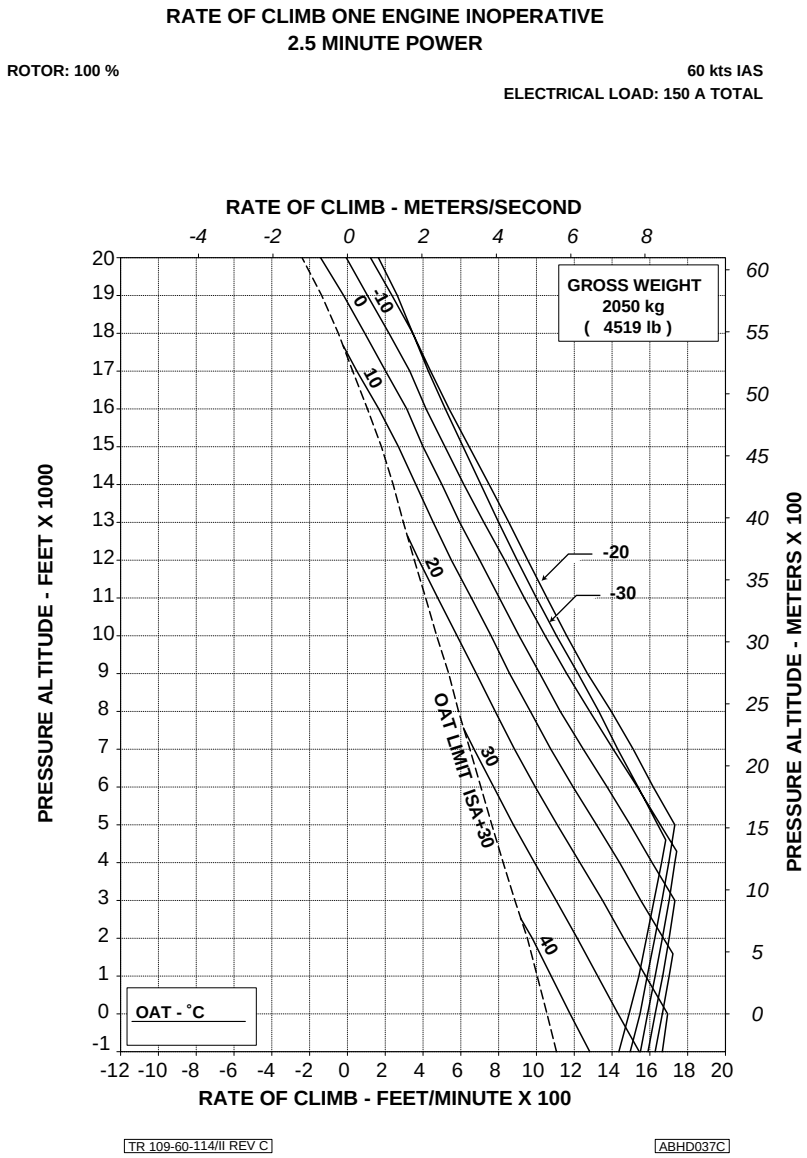

Figure 4-17. Rate of climb - OEI - 2.5 minutes power - 2050 kg.

FOR TRAINING ONLY

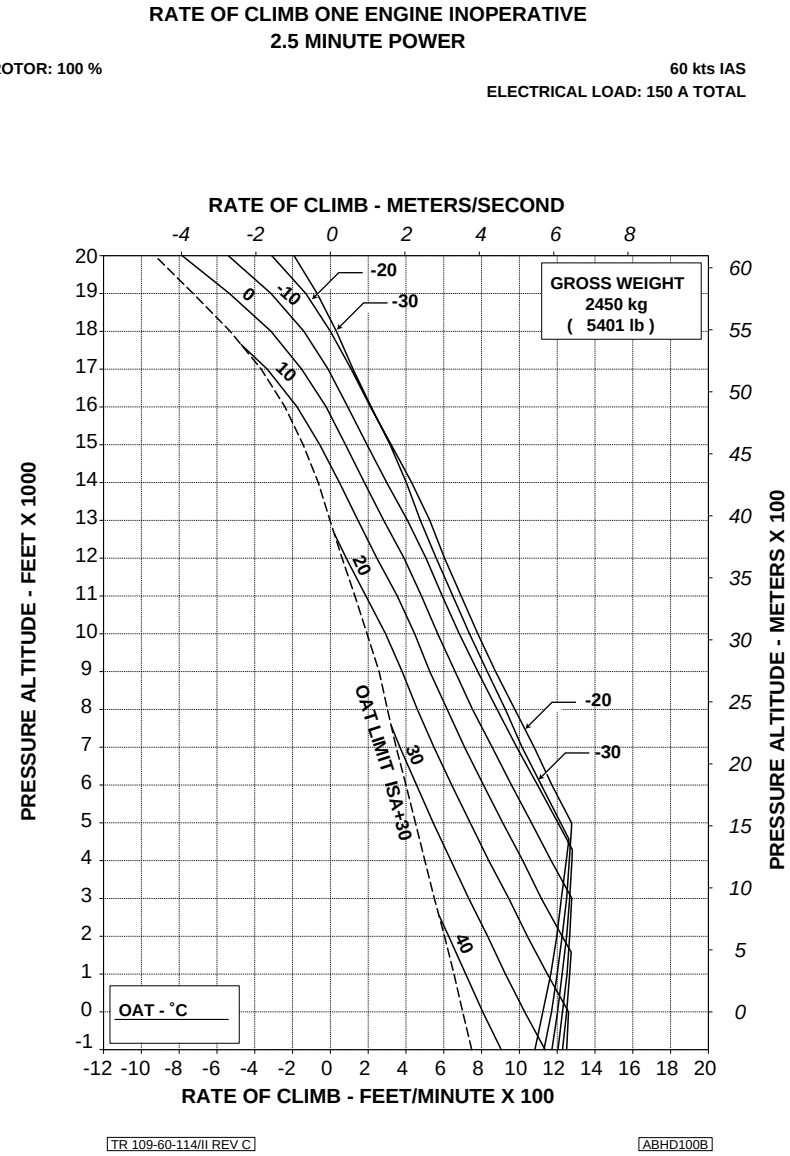

Figure 4-18. Rate of climb - OEI - 2.5 minutes power - 2450 kg.

FOR TRAINING ONLY

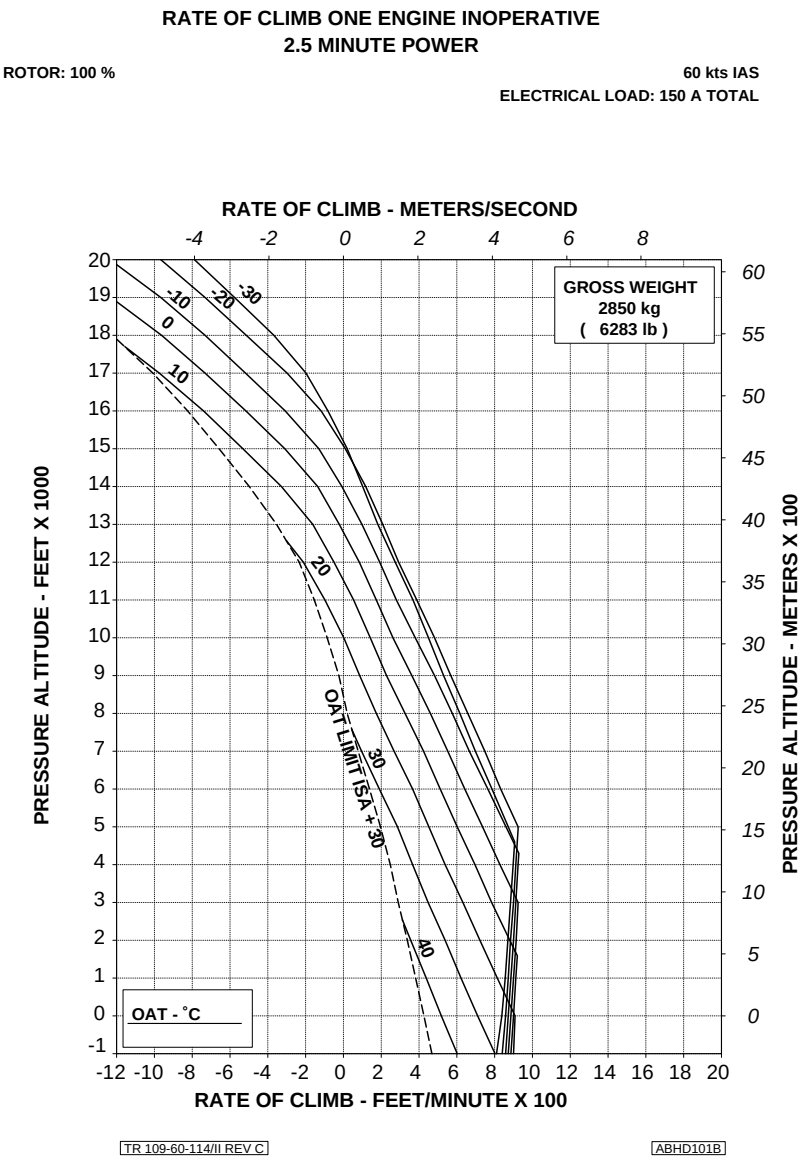

Figure 4-19. Rate of climb - OEI - 2.5 minutes power - 2850 kg.

FOR TRAINING ONLY

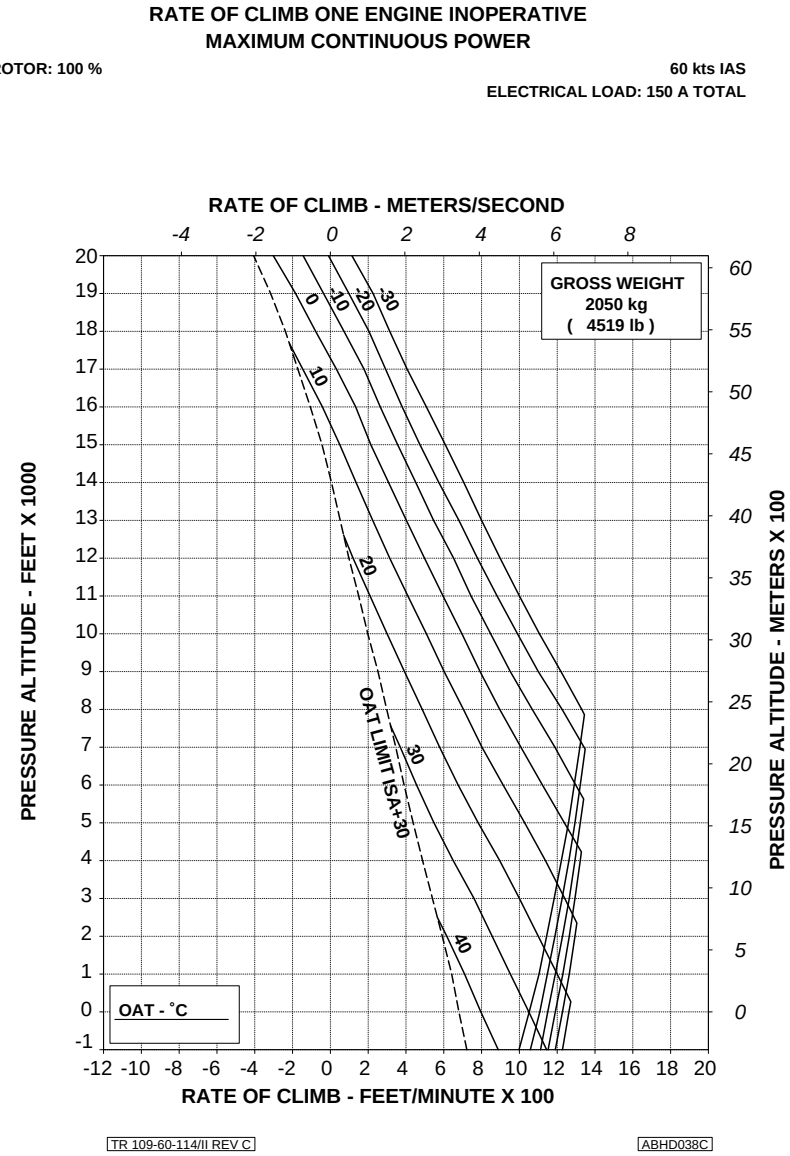

Figure 4-20. Rate of climb - OEI - maximum continuous power - 2050 kg.

FOR TRAINING ONLY

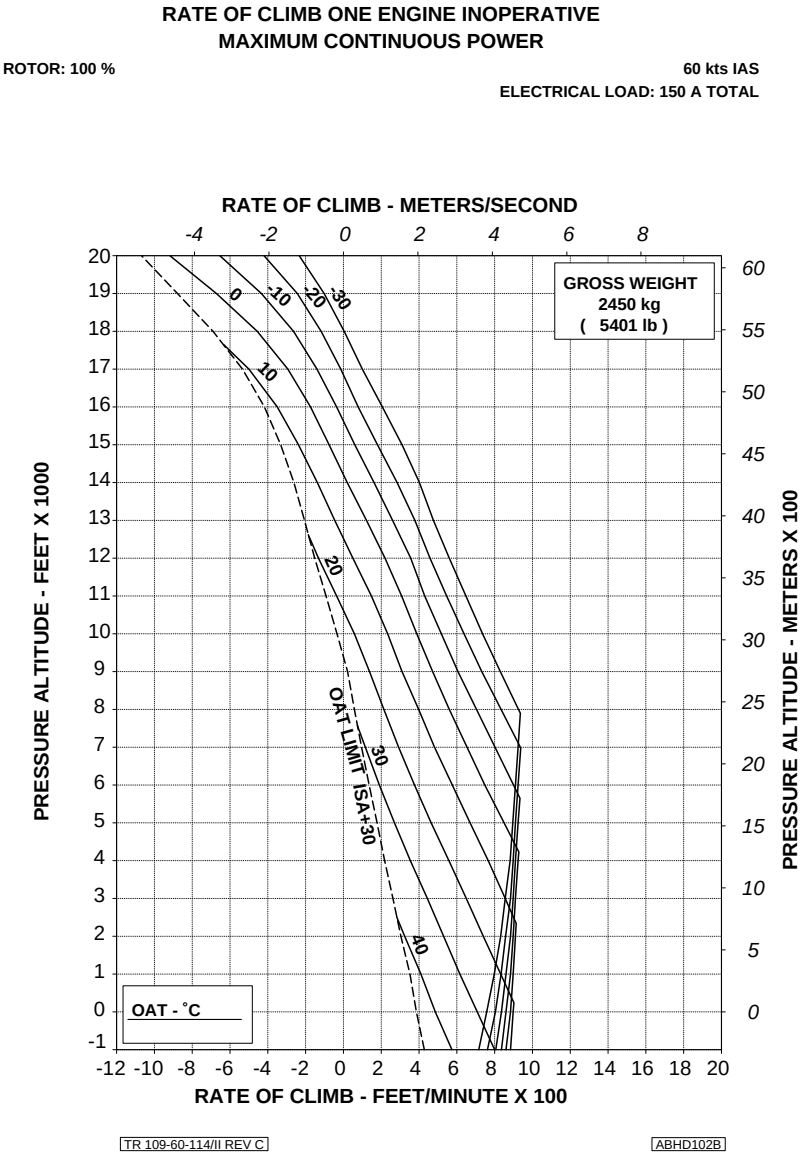

Figure 4-21. Rate of climb - OEI - maximum continuous power - 2450 kg.

FOR TRAINING ONLY

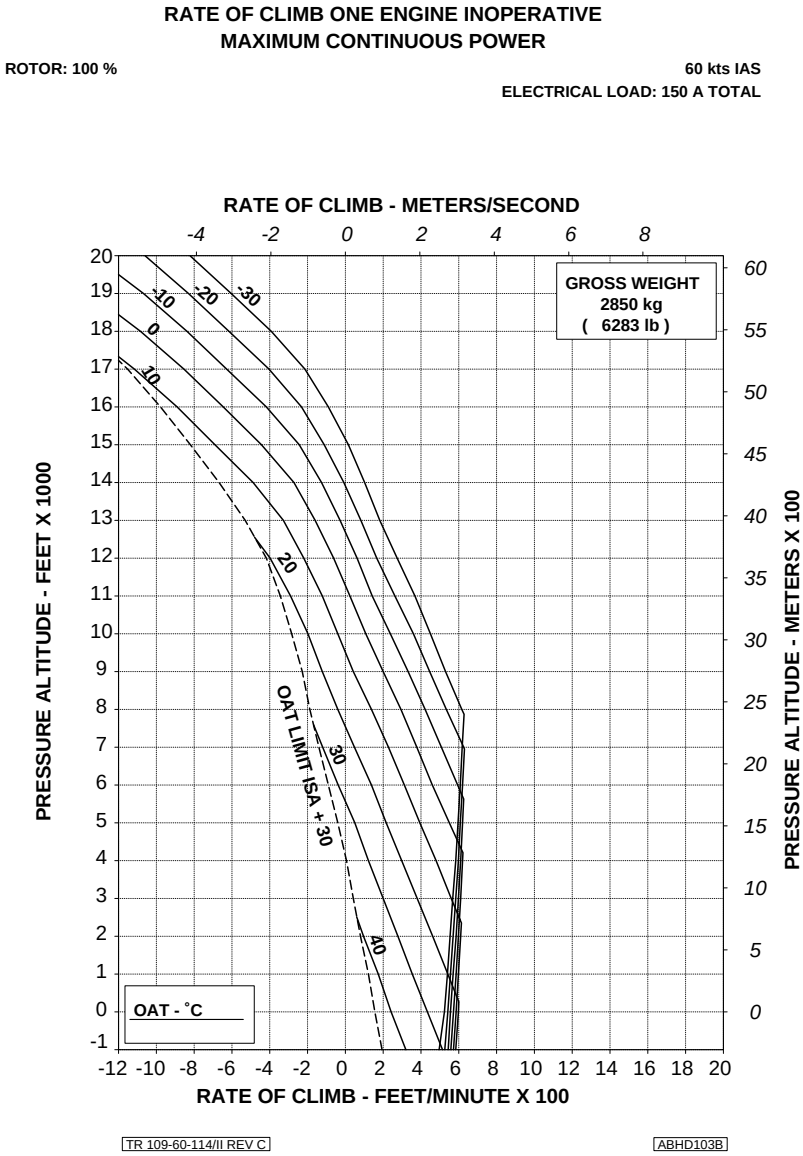

Figure 4-22. Rate of climb - OEI - maximum continuous power - 2850 kg.

FOR TRAINING ONLY

AIRSPEED CALIBRATION

The Airspeed Calibration charts provide calibrated airspeed in forward flight for pilot and copilot airspeed systems.

FOR TRAINING ONLY

AIRSPEED CALIBRATION CURVE
PILOT
(FORWARD FLIGHT)

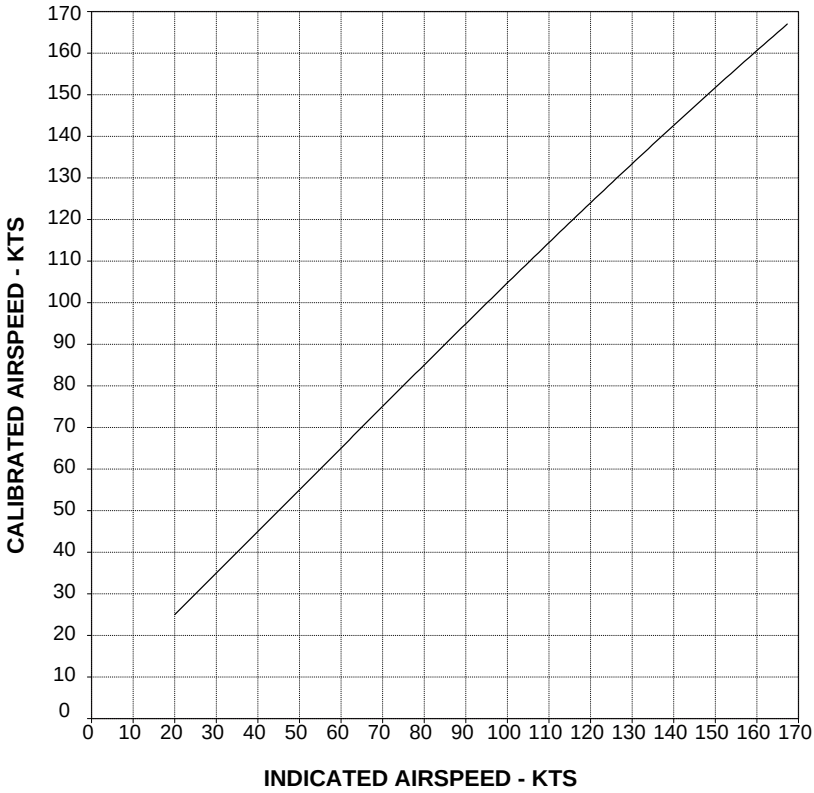

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Figure 4-23. Airspeed calibration curve pilot (forward flight).

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AIRSPEED CALIBRATION CURVE
COPILOT
(FORWARD FLIGHT)

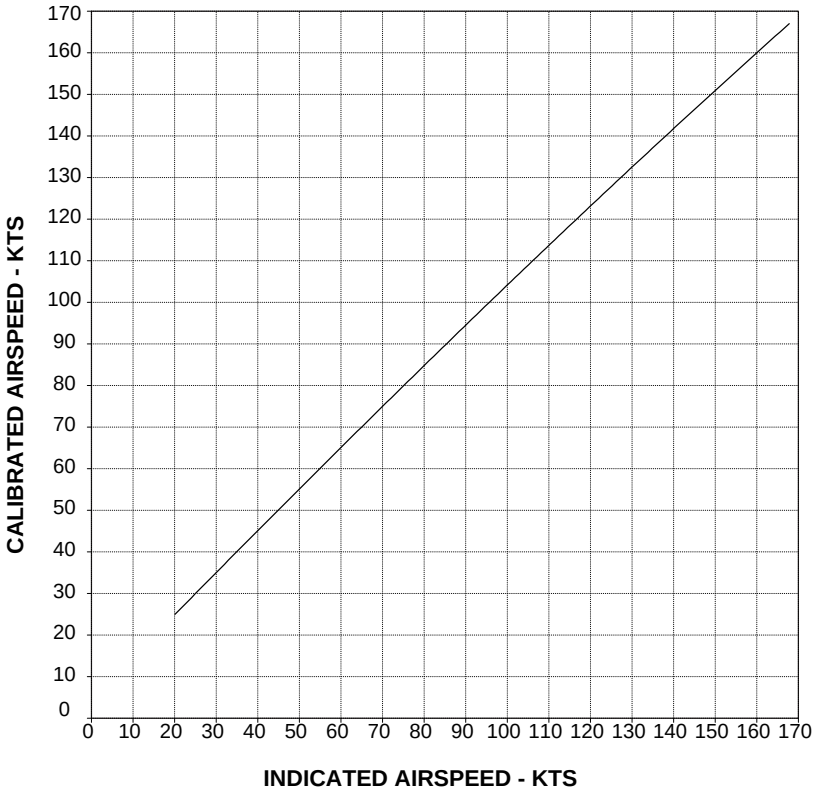

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Figure 4-24. Airspeed calibration curve copilot (forward flight).

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